Truth lemma for the canonical model

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the tagitem/iftag conditional blocks carry Exercise tags in the active group and the complete argument in the else group - rendered in full once the preamble iftag branch semantics is fixed; depends on prop:ccs-properties (registered card L42016-lineage sibling) as a stub.

Editorial note. The label prop:truthlemma names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Canonical model satisfaction clauses. Modal logic syntax and truth unit, defn:mmodels. Each inductive case converts mSat into the clause-wise form.
- Closure properties of maximal consistent sets. Same unit, \ollabel{prop:ccs-properties} with sub-labels prop:ccs-lfalse / prop:ccs-ltrue / prop:ccs-lnot / prop:ccs-land / prop:ccs-lor / prop:ccs-lif (registered card). Membership side of every biconditional: the MCS closure facts for each connective.

Source reference. Original proposition prop-ccs-land: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 489–489. Prerequisite pointer invoked at proof line(s) 1372; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-lfalse: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 482–482. Prerequisite pointer invoked at proof line(s) 1347; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-lif: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 493–493. Prerequisite pointer invoked at proof line(s) 1390; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-liff: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 495–495. Prerequisite pointer invoked at proof line(s) 1402; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-lnot: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 486–486. Prerequisite pointer invoked at proof line(s) 1363; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-lor: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 491–491. Prerequisite pointer invoked at proof line(s) 1381; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-ltrue: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 484–484. Prerequisite pointer invoked at proof line(s) 1351; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-properties: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 475–475. Prerequisite pointer invoked at proof line(s) 1347; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-box-iff-diamond: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1247–1247. Prerequisite pointer invoked at proof line(s) 1445; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-box: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1185–1185. Prerequisite pointer invoked at proof line(s) 1413; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-diamond: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1277–1277. Prerequisite pointer invoked at proof line(s) 1436; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-mmodels: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1345–1345. Named citation invoked at source line(s) 1345: the author's marker \olref[syn][trw]{defn:mmodels} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1 (Truth Lemma)**.** *For every formula A, $M^\Sigma \models A[]\Delta]$ if and only if $A \in \Delta$.*

Proof. By induction on A.

1. $A \equiv \bot$: $\not\models M^\Sigma[\bot][\Delta]$ by defn-mmodels, and $\bot \notin \Delta$ by prop-ccs-propertiesprop-ccs-lfalse.
2. $A \equiv \top$: $M^\Sigma \models \top[]\Delta]$ by defn-mmodels, and $\top \in \Delta$ by prop-ccs-propertiesprop-ccs-ltrue.
3. $A \equiv p$: $M^\Sigma \models p[]\Delta]$ iff $\Delta \in V^\Sigma(p)$ by defn-mmodels. Also, $\Delta \in V^\Sigma(p)$ iff $p \in \Delta$ by definition of V^Σ.
4. $A \equiv \neg B$: Exercise.
5. $A \equiv B \wedge C$: Exercise.
6. $A \equiv B \vee C$: Exercise.
7. $A \equiv B \to C$: Exercise.
8. $A \equiv B \leftrightarrow C$: Exercise.
9. $A \equiv \Box B$: Exercise.
10. $A \equiv \Diamond B$: Exercise.

□